

PIA SWARUP

(781) 250-9382 | pswarup@olin.edu | Needham, MA, USA | www.linkedin.com/in/piaswarup

EDUCATION

Olin College of Engineering, Needham, MA | Bachelor of Science in Mechanical Engineering May 2029

- Relevant coursework: Mechanical Prototyping, Software Design, Modeling & Simulation, Linear Algebra, Quantitative Engineering Analysis, Instrumentation & Measurement, Biomimicry, Products & Markets

SKILLS

Software: SolidWorks, Onshape, Autodesk Fusion, MATLAB, HTML/CSS, Python, Emulate3D

Design: Illustrator, Photoshop, WordPress, Microsoft Office, Canva

Fabrication: TIG & MIG Welder, Manual and CNC Mill, CNC Router, FDM and Resin 3D Printer, Saws, Soldering, Manual and CNC Lathe, Laser Cutter, Plasma Cutter, Waterjet

Language: Dutch (Advanced), German (Advanced), French (Basic)

EXPERIENCE

Robotics Intern | Movu NV, Belgium Feb 2024 - July 2025

- Developed 3D simulations for AGVs and robot picking arms, improving traffic control, throughput optimization and reducing design iteration time

Web Design and Marketing Intern | Veg Nutraceuticals sp. z o.o., Poland Jan 2022 - Jan 2026

- Web design & E-commerce with WordPress and Shopify
- Devised Manual Labelling Apparatus
- Designed labels and posters, made product mockups
- Marketing and SEO on Instagram, Facebook, and Google

Teaching Assistant | Codefever, Belgium Nov 2024 - Jan 2025

- Taught coding concepts and computational thinking to primary and secondary school students

PROJECTS & EXTRACURRICULAR

High School VEX Robotics Team | Founder & President Sep 2023

- Top 10 in 2024 & 2025 VEX Robotics Benelux Championships
- Led robot design and assembly, including chassis and mechanical systems, to ensure competitive performance
- Founded and grew a sustainable robotics team by leading fundraising, budgeting, technical documentation, and team infrastructure

Human-centered Artificial Intelligence and Robotics Lab | Mechanical Engineer Feb 2026

- Developing a drone-mounted soil sampling robot to measure soil moisture in agricultural environments
- Designing base structure and anchoring mechanisms to ensure stability and reliable ground interaction

Olin Formula SAE & Olin Baja SAE | Suspension Engineer Sep 2025

- Designing and analyzing the rear Knuckle for Formula Electric Racing Car
- Designed Wheel Hubs focusing on Design for Manufacturing and performed FEA for Baja Off-road vehicle
- Performing Suspension Kinematics Analysis and Simulation

Autonomous Self-Balancing and Self-Driving Bike Feb 2026

- Developing a self-balancing, driverless bicycle capable of autonomous indoor/outdoor navigation